

Continuum Mechanics

class #04

8/27/2026

Eigenvalues, Eigenvectors and Principal Invariants

Def: By an eigenpair for a second order tensor $S \in V^2$ we mean a scalar, $\lambda \in \mathbb{R}$, and a unit vector, $\vec{e} \in V$, satisfying

$$S\vec{e} = \lambda\vec{e}$$

Any such λ is called an eigenvalue and any such $\vec{e}$ is called an eigenvector of S .

Proposition: let $S \in V^2$ be given. If $\lambda \in \mathbb{R}$ is an eigenvalue of S then $p(\lambda) = 0$, where

$$(1.15) \quad p(\lambda) = \det(S - \lambda I).$$

p is called the characteristic polynomial.

Proposition: Suppose that $S \in V^2$ is given and $\lambda \in \mathbb{R}$ is an eigenvalue of S . A solution to the equation

$$(1.16) \quad (S - \lambda I)\vec{e} = \vec{0}$$

yields non-unique eigenvectors of S associated to λ .

Remark: Here we have decided to ignore complex eigenvalues, not because they are unimportant, but because they are not as useful in Continuum Mechanics. Note that solutions to

$$p(\lambda) = 0$$

could come in complex-conjugate pairs.

Result (1.8): (1) The eigenvalues of SPD tensors are strictly positive. (2) Any two eigenvectors corresponding to distinct eigenvalues of a symmetric tensor are orthogonal.

Proof: (1) Let $S \in V^2$ be SPD and let $(\lambda, \vec{e})$ be an eigenpair. Then,

$$S\vec{e} = \lambda\vec{e}$$

Thus,

$$0 < \vec{e} \cdot (S\vec{e}) = \vec{e} \cdot (\lambda\vec{e}) = \lambda (\vec{e} \cdot \vec{e}) = \lambda. //$$

(2) Suppose that S is symmetric and $(\lambda, \vec{w})$ and $(\mu, \vec{u})$ be eigenpairs. Assume $\lambda \neq \mu$. Then

$$\lambda (\vec{u} \cdot \vec{w}) = \vec{u} \cdot (\lambda \vec{w})$$

$$= \vec{u} \cdot (S\vec{w})$$

$$= \vec{w} \cdot (S\vec{u}) \quad (\text{symmetry})$$

$$= \vec{w} \cdot (\mu \vec{u})$$

$$= \mu (\vec{w} \cdot \vec{u})$$

Hence

$$(\lambda - \mu)(\vec{w} \cdot \vec{w}) = 0.$$

This implies, by cancellation, that

$$\vec{w} \cdot \vec{w} = 0. \quad ///$$

Result (1.9) (Spectral Decomposition Theorem) Let $S \in V^2$ be symmetric. There is a right-handed orthonormal basis (frame) $\mathcal{F} = \{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ consisting of eigenvectors of S . The corresponding eigenvalues λ_i , $i=1,2,3$, are the same (up to ordering) for any such basis and the full set of eigenvalues of S . The dyadic representation of S in any such basis is

$$S = \sum_{i=1}^3 \lambda_i \vec{e}_i \otimes \vec{e}_i$$

and the matrix representation is

$$[S] = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}.$$

Proof: A good book on linear algebra. ///

Definition: Let $S \in V^2$ be arbitrary. The principal invariants of S are the 3 scalar quantities

$$I_1(S) = \text{tr}(S)$$

$$I_2(S) = \frac{1}{2} \left[(\text{tr}(S))^2 - \text{tr}(S^2) \right]$$

and

$$I_3(S) = \det(S)$$

Proposition: let $S \in V^2$ be a given tensor and $\alpha \in \mathbb{R}$ be any scalar. Then

$$\begin{aligned} \det(S - \alpha I) &= -\alpha^3 + I_1(S)\alpha^2 - I_2(S)\alpha + I_3(S) \\ &= p(\alpha) \end{aligned}$$

Proof: let $\mathcal{H} = \{\vec{e}_i\}$ be a frame. Compute the characteristic polynomial $p(\alpha)$ and compare coefficients. ///

Proposition: let $S \in V^2$ be symmetric. Then

$$I_1(S) = \lambda_1 + \lambda_2 + \lambda_3$$

$$I_2(S) = \lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1$$

$$I_3(S) = \lambda_1\lambda_2\lambda_3$$

Proof: Exercise. ///

$$I_1, I_2, I_3$$

Remark: The invariants are so-called because they are built from the invariants $\text{tr}(S)$ and $\det(S)$. See Results 1.5 and 1.6. //

Result (1.10) (Cayley-Hamilton Theorem) let $S \in V^2$ be arbitrary. Then

$$S^3 - I_1(S)S^2 + I_2(S)S + I_3(S)I = 0$$

Proof: Exercise. ///

Special Decomposition

Result (1.11): let $C \in V^2$ be SPD. There is a unique SPD tensor U such that $U^2 = C$.
In particular,

$$U = \sum_{i=1}^3 \sqrt{\lambda_i} \vec{e}_i \otimes \vec{e}_i$$

where $\mathcal{H} = \{\vec{e}_i\}$ is any frame of eigenvectors of C and $\lambda_i > 0$ are the corresponding eigenvalues of C .

Proof: Exercise. ///

Result (1.12): let $F \in V^2$ be any tensor satisfying

$$\det(F) > 0.$$

Then there exist right and left polar decompositions of F taking the forms, respectively,

$$F = RU$$

$$F = VR$$

where

$$U = \sqrt{F^T F} \quad \text{and} \quad V = \sqrt{F F^T},$$

which are both SPD, and R is a rotation.

Proof: Since $\det(F) > 0$, F is invertible. Thus

$$F\vec{v} \neq \vec{0} \quad \forall \vec{v} \in \mathcal{V}_* := \mathcal{V}_* - \{\vec{0}\}.$$

Similarly, F^T is invertible and

$$F^T \vec{v} \neq \vec{0} \quad \forall \vec{v} \in \mathcal{V}_*$$

Thus

$$\vec{v} \cdot (F^T F \vec{v}) = (F\vec{v}) \cdot (F\vec{v}) > 0, \quad \forall \vec{v} \in \mathcal{V}_*,$$

$$\vec{v} \cdot (F F^T \vec{v}) = (F^T \vec{v}) \cdot (F^T \vec{v}) > 0, \quad \forall \vec{v} \in \mathcal{V}_*.$$

Thus $F F^T$ and $F^T F$ are SPD. Using Result 1.11, define

$$U := \sqrt{F^T F} \quad V := \sqrt{F F^T}$$

Next, define

$$R := F U^{-1}$$

Observe,

$$R U = F$$

$$0 < \det(F)$$

$$= \det(R U)$$

$$= \det(R) \det(U)$$

Also note that

$$\det(U) > 0$$

Since U is SPD. So

$$\det(R) > 0.$$

Furthermore,

$$R^T R = U^{-T} F^T F U^{-1} = U^{-1} U^2 U^{-1} = I$$

So R is orthogonal and

$$|\det(R)| = 1$$

Clearly $\det(R) = 1$ and R is a rotation. The right polar decomposition

$$F = RU$$

is proven. To get the left polar decomposition, define

$$Q := V^{-1}F$$

Use the same technique as above to show that Q is a rotation. So we have

$$RU = F = VQ.$$

let

$$S := Q^T V Q$$

Then S is SPD. (show this.)

And

$$\begin{aligned}RU &= VQ \\&= QQ^T VQ \\&= QS,\end{aligned}$$

which implies

$$R = QS U^{-1}$$

Since R is a rotation, $R^{-1} = R^T$, which implies

$$S^2 = U^2 = F^T F$$

$\uparrow$
 $U = \sqrt{F^T F}$

By uniqueness of square roots $U = S$. Then

$$\begin{aligned}R &= QS U^{-1} \\&= Q.\end{aligned}$$

Therefore

$$RU = F = VR \quad ///$$

Scalar Product for Second-Order Tensors

Definition : Let $S, D \in V^2$. Then

(1.17)

$$S:D := \text{tr}(S^T D)$$

defines what is called the tensor inner product (for second-order tensors).

Proposition: let $\mathcal{F} = \{\vec{e}_i\}$ be a frame. Then the dyadic basis for V^2 , defined as the set

$$\{\vec{e}_i \otimes \vec{e}_j \mid 1 \leq i, j \leq 3\}$$

is orthonormal with respect to the tensor inner product, i.e.

$$(\vec{e}_i \otimes \vec{e}_j) : (\vec{e}_k \otimes \vec{e}_l) = \delta_{ik} \delta_{jl}$$

Proof: Exercise 1.28. $\equiv$

Result (1.13): let $\mathcal{F} = \{\vec{e}_i\}$ be a frame. Suppose

$$S = S_{ij} \vec{e}_i \otimes \vec{e}_j$$

$$D = D_{ij} \vec{e}_i \otimes \vec{e}_j$$

Then

$$S : D = S_{ij} D_{ij}$$

If S is symmetric, then

$$S : D = S : \text{Sym}(D)$$

$$= \text{sym}(S) : \text{Sym}(D)$$

Proof: Add proof in slides. //

Fourth-Order Tensors

Definition: A fourth-order tensor is a mapping $C: V^2 \rightarrow V^2$ that is linear in the sense that

$$1) \quad C(S+T) = CS + CT \quad \forall S, T \in V^2$$

$$2) \quad C(\alpha T) = \alpha CT \quad \forall \alpha \in \mathbb{R} \quad \forall T \in V^2.$$

The set of all fourth-order tensors is denoted V^4 .
The zero four-tensor, $0 \in V^4$, is the tensor with the property that

$$0S = 0 \in V^2, \quad \forall S \in V^2.$$

The identity four-tensor, $I \in V^4$, is the tensor with the property that

$$IS = S, \quad \forall S \in V^2$$

Two fourth-order tensors $C, D \in V^4$ are called equal iff

$$CT = DT, \quad \forall T \in V^2.$$

Example: Fix $A \in V^2$. Define

$$C(T) = AT, \quad \forall T \in V^2$$

Then C is a four tensor.

Typeset
Note

$C \in V^4$
should be
boldface
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Proposition: let $C, D \in V^4$ and $\alpha, \beta \in \mathbb{R}$. Define

$$(\alpha C + \beta D)^T := \alpha C(T) + \beta D(T)$$

for all $T \in V^2$. Define the product $CD: V^2 \rightarrow V^2$ via

$$CD(T) = C(D(T)) \quad \forall T \in V^2.$$

Then V^4 is a linear vector space. Furthermore, $CD \in V^4$

Proof: An easy, if exhausting exercise. ///

Representation in a Coordinate Frame

Definition: let $C \in V^4$ and suppose $\mathcal{F} = \{\vec{e}_i\}$ is a right-handed, orthonormal frame. The 81 components of C ,

$$C_{ijkl} \in \mathbb{R}, \quad 1 \leq i, j, k, l \leq 3,$$

are defined via

$$C_{ijkl} = e_i \cdot (C(\vec{e}_k \otimes \vec{e}_l) \vec{e}_j).$$

Proposition: Suppose that $C \in V^4$ and $U, T \in V^2$ satisfy

$$U = C(T).$$

let $\mathcal{F} = \{\vec{e}_i\}$ be a frame and suppose

$$U = U_{ij} \vec{e}_i \otimes \vec{e}_j \quad T = T_{ij} \vec{e}_i \otimes \vec{e}_j$$

Then

$$(1.18) \quad U_{ij} = C_{ijkl} T_{kl}.$$

Proof: We have

$$\begin{aligned} U_{ij} &= \vec{e}_i \cdot (U \vec{e}_j) \\ &= \vec{e}_i \cdot (C(T) \vec{e}_j) \\ &= \vec{e}_i \cdot (C(T_{kl} \vec{e}_k \otimes \vec{e}_l) \vec{e}_j) \\ &= \vec{e}_i \cdot (C(\vec{e}_k \otimes \vec{e}_l) \vec{e}_j) T_{kl} \\ &= C_{ijkl} T_{kl} \quad // \end{aligned}$$

Example: let $A \in V^2$ be given and define $C \in V^4$ by

$$C(T) = AT$$

Then

$$\begin{aligned} C_{ijkl} &= \vec{e}_i \cdot (C(\vec{e}_k \otimes \vec{e}_l) \vec{e}_j) \\ &= \vec{e}_i \cdot ((A \vec{e}_k \otimes \vec{e}_l) \vec{e}_j) \\ &= \vec{e}_i \cdot (A(\vec{e}_k \otimes \vec{e}_l) \vec{e}_j) \\ &= \vec{e}_i \cdot (A(\vec{e}_l \cdot \vec{e}_j) \vec{e}_k) \\ &= \vec{e}_i \cdot (A \delta_{lj} \vec{e}_k) \end{aligned}$$

$$= \delta_{ej} \vec{e}_i \cdot (A \vec{e}_k)$$

$$= \delta_{ej} A_{ik}$$

$$= A_{ik} \delta_{ej} //$$

Proposition: let $C \in V^4$. Then

$$C_{ijkl} = (\vec{e}_i \otimes \vec{e}_j) : C (\vec{e}_k \otimes \vec{e}_l)$$

Proof: We want to show that

$$\vec{e}_i \cdot (C (\vec{e}_k \otimes \vec{e}_l) \vec{e}_j) = (\vec{e}_i \otimes \vec{e}_j) : C (\vec{e}_k \otimes \vec{e}_l)$$

In general, we have, for any $A \in V^2$,

$$A_{ij} = \vec{e}_i \cdot (A \vec{e}_j) = \vec{e}_i \otimes \vec{e}_j : A.$$

To see this, observe that

$$\vec{e}_i \otimes \vec{e}_j = \delta_{ik} \delta_{jl} \vec{e}_k \otimes \vec{e}_l \quad (\text{sum on } k, l)$$

Thus, using Result 1.13, for fixed i, j ,

$$\begin{aligned} \vec{e}_i \otimes \vec{e}_j : A &= \delta_{ik} \delta_{jl} A_{kl} \\ &= A_{ij} \end{aligned}$$

Using

$$\vec{e}_i \cdot (A^{(k,l)} \vec{e}_j) = \vec{e}_i \otimes \vec{e}_j : A^{(k,l)}$$

with

$$A^{(k,l)} := C(\vec{e}_k \otimes \vec{e}_l),$$

proves the result.